

Machine Learning

PENG

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1 Linear Algebra

Definition 1.1. *Linear dependency:*

$$A_{n \times n} x_n = 0$$
$$\begin{bmatrix} \vdots \\ a_1 \\ \vdots \end{bmatrix} x_1 + \begin{bmatrix} \vdots \\ a_2 \\ \vdots \end{bmatrix} x_2 + \begin{bmatrix} \vdots \\ a_n \\ \vdots \end{bmatrix} x_n = 0$$

If $a_1, \dots, a_n$ is linearly independent $\Rightarrow x_1 = \dots = x_n = 0$

If $a_1, \dots, a_n$ is linearly dependent $\Leftrightarrow \det(A) = 0 \Rightarrow \text{nonzero solution}$

Definition 1.2. *Eigenvalue:*

$$A_{n \times n} x_n = \lambda x_{n \times 1} \quad (1.1)$$

$$(A - \lambda I)x = 0$$

$$|A - \lambda I| = 0 \quad (1.2)$$

Definition 1.3. *Orthogonal:*

2 Singular Value Decomposition

Definition 2.1. *SVD [1]:*

$$X_{n \times m} = U_{n \times n} \Sigma_{n \times m} V_{m \times m}^T \quad (2.1)$$

where

$$UU^T = U^T U = I_{n \times n}$$

$$VV^T = V^T V = I_{m \times m}$$

$$\Sigma_{n \times m} = \begin{bmatrix} \lambda_1 > 0 & & & \\ & \ddots & & \\ & & \lambda_m > \lambda_{m-1} & \\ & & \vdots & \end{bmatrix}$$

Definition 2.2. *Economy SVD ($n \geq m$):*

$$X_{n \times m} = \begin{bmatrix} \hat{U}_{n \times m} & \hat{U}_{n \times (n-m)}^\perp \end{bmatrix} \begin{bmatrix} \hat{\Sigma}_{m \times m} \\ 0_{(n-m) \times m} \end{bmatrix} V_{m \times m}^T \quad (2.2)$$
$$= \hat{U}_{n \times m} \hat{\Sigma}_{m \times m} V_{m \times m}^T$$

Example 2.3.

$$A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$$

1. $AA^T = \begin{bmatrix} 17 & 8 \\ 8 & 17 \end{bmatrix}$

2. *Eigenvalues of AA^T*

$$\begin{aligned} |AA^T - \lambda I| &= 0 \\ \begin{vmatrix} 17 - \lambda & 8 \\ 8 & 17 - \lambda \end{vmatrix} &= 0 \\ (\lambda - 25)(\lambda - 9) &= 0 \end{aligned}$$

$$\lambda_1 = 9, \lambda_2 = 25$$

3. *Right singular vectors (Eigenvectors of AA^T)*

For $\lambda_1 = 9$:

$$\begin{aligned} (AA^T - \lambda_1 I)v &= 0 \\ \begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix} v &= 0 \end{aligned}$$

References

- [1] Steven L Brunton and J Nathan Kutz. *Data-driven science and engineering: Machine learning, dynamical systems, and control*. Cambridge University Press, 2022. ISBN: 1009098489.